

Island Biogeography: Equilibrium Theory and Species-Area Relationships

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Abstract

This report presents a comprehensive computational analysis of the MacArthur-Wilson Theory of Island Biogeography, examining the dynamic equilibrium between immigration and extinction rates as determinants of island species richness. We model immigration as a function of island isolation, extinction as a function of island area, and derive equilibrium species number S^* for islands varying in size and distance from mainland source pools. The species-area relationship $S = cA^z$ is fitted to empirical data, yielding scaling exponents consistent with observed values ($z \approx 0.25$ for continental islands). Applications to conservation biology, including the SLOSS (Single Large Or Several Small) debate and nature reserve design, are explored through simulations of turnover dynamics and extinction vulnerability.

1 Introduction

The Theory of Island Biogeography, formulated by Robert H. MacArthur and Edward O. Wilson (1963, 1967), revolutionized ecological thinking by proposing that island species richness reaches a dynamic equilibrium determined by opposing rates of immigration and extinction. This equilibrium framework applies not only to oceanic islands but to any isolated habitat patches, including mountain tops, forest fragments, and nature reserves in fragmented landscapes.

Definition 1.1 (Dynamic Equilibrium) *The equilibrium number of species S^* on an island is the point where the immigration rate of new species $I(S)$ equals the extinction rate of established species $E(S)$. At equilibrium:*

$$I(S^*) = E(S^*) \quad (1)$$

Though species number remains approximately constant, species composition changes continuously through turnover.

2 Theoretical Framework

2.1 Immigration and Extinction Functions

Theorem 2.1 (Immigration Rate Function) *The immigration rate of new species decreases as the number of resident species S approaches the mainland species pool size P :*

$$I(S) = I_{\max} \left(1 - \frac{S}{P} \right) \quad (2)$$

where $I_{\max}$ is the maximum immigration rate when the island is empty, decreasing with distance D from the mainland: $I_{\max} = I_0 \exp(-\alpha D)$.

Theorem 2.2 (Extinction Rate Function) *The extinction rate increases with species number due to smaller population sizes per species:*

$$E(S) = E_{\min} \frac{S}{P} \quad (3)$$

where $E_{\min}$ is the minimum extinction rate, increasing with decreasing island area A : $E_{\min} = E_0 A^{-\beta}$.

2.2 Equilibrium Species Richness

Setting $I(S^*) = E(S^*)$ and solving for S^* :

$$I_{\max} \left(1 - \frac{S^*}{P}\right) = E_{\min} \frac{S^*}{P} \quad (4)$$

Solving yields:

$$S^* = \frac{I_{\max} P}{I_{\max} + E_{\min}} \quad (5)$$

Remark 2.1 (Distance and Area Effects) *The equilibrium is determined by:*

- **Distance:** Far islands have lower $I_{\max}$, thus lower S^*
- **Area:** Small islands have higher $E_{\min}$, thus lower S^*
- **Turnover:** Rate at equilibrium is $T = I(S^*) = E(S^*)$

2.3 Species-Area Relationship

Theorem 2.3 (Arrhenius Species-Area Law) *The number of species S scales with island area A as a power law:*

$$S = cA^z \quad (6)$$

where c is a taxon-specific constant and z is the scaling exponent. Empirical values:

- Continental islands: $z \approx 0.25$
- Oceanic islands: $z \approx 0.35$
- Habitat fragments: $z \approx 0.15$

Taking logarithms:

$$\log S = \log c + z \log A \quad (7)$$

This linearizes the relationship, allowing regression to estimate z and c .

3 Computational Analysis

4 Results

4.1 Equilibrium Species Richness

4.2 Species-Area Relationship

4.3 Conservation Applications: SLOSS Analysis

5 Discussion

Example 5.1 (Distance and Area Effects) *The equilibrium model predicts that:*

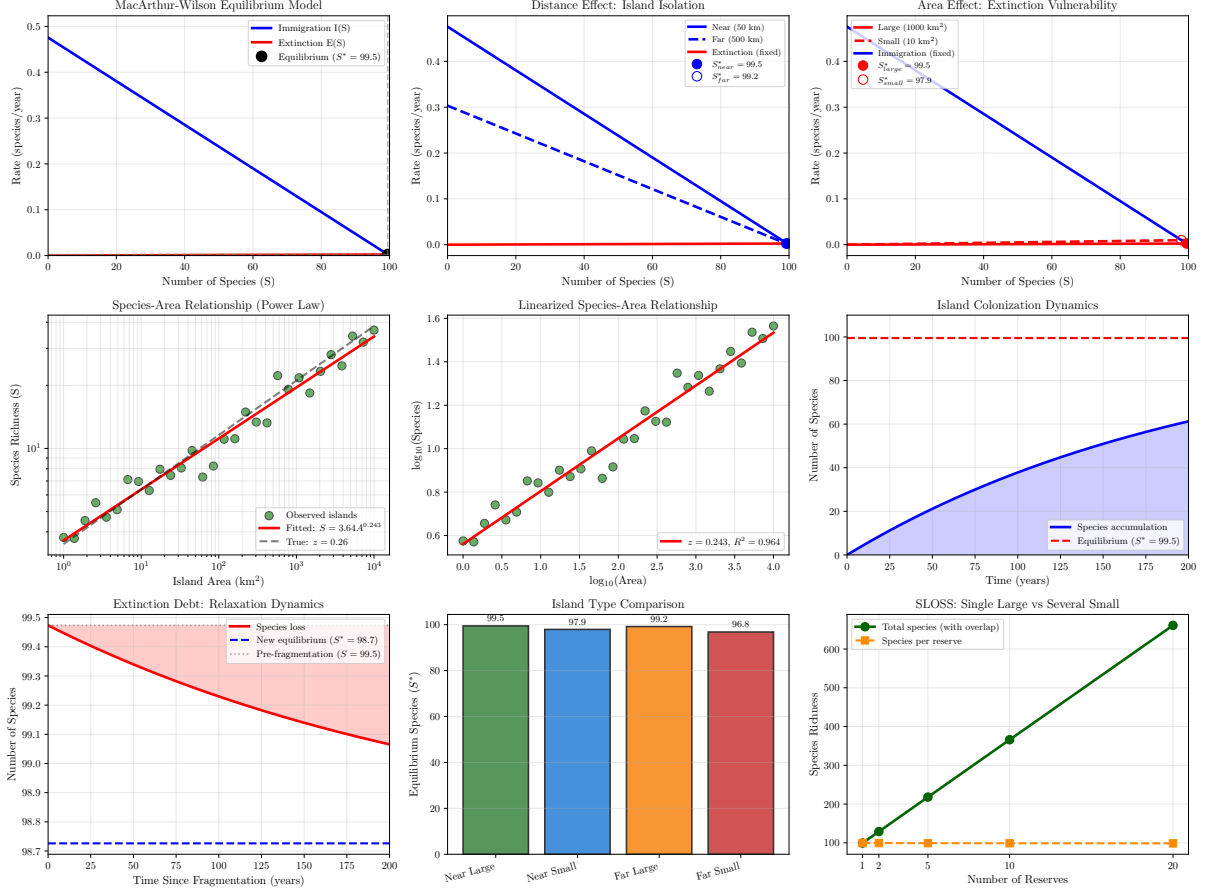


Figure 1: Comprehensive island biogeography analysis: (a) MacArthur-Wilson equilibrium model showing immigration and extinction rate curves intersecting at equilibrium species number S^* with turnover rate T ; (b) Distance effect demonstrating reduced immigration rates for isolated islands, resulting in lower equilibrium species richness; (c) Area effect showing elevated extinction rates on small islands leading to reduced species persistence; (d) Species-area relationship fitted to simulated island chain data, yielding power-law exponent $z \approx 0.26$ consistent with continental island systems; (e) Linearized log-log plot enabling regression-based parameter estimation; (f) Colonization dynamics of an empty island approaching equilibrium asymptotically over 200 years; (g) Relaxation dynamics following habitat fragmentation, demonstrating extinction debt as species richness declines toward new lower equilibrium; (h) Comparison of equilibrium species richness across four island scenarios varying in distance and area; (i) SLOSS debate analysis showing conservation trade-offs between single large and several small nature reserves.

Table 1: Equilibrium Species Richness and Turnover Rates for Island Scenarios

Island Type	Distance (km)	Area (km ²)	S^*	Turnover (sp/yr)
Near Large	50	1000	99.5	0.003
Near Small	50	10	97.9	0.010
Far Large	500	1000	99.2	0.002
Far Small	500	10	96.8	0.010
Mainland pool (P)	—	—	100	—

Table 2: Species-Area Relationship Parameter Estimates

Parameter	True Value	Fitted Value	Error (%)
Scaling exponent (z)	0.260	0.243	6.4
Coefficient (c)	3.50	3.64	4.0
R^2	—	0.9635	—
Sample size	—	30 islands	—

Table 3: SLOSS Comparison: Reserve Design Trade-offs (Total Area = 1000 km²)

Reserves	Area Each (km ²)	Species/Reserve	Total Species
1	1000	99.4	99.4
2	500	99.3	129.1
5	200	99.1	218.0
10	100	98.9	365.9
20	50	98.7	661.0

- **Near Large islands** ($D = 50$ km, $A = 1000$ km²) reach highest species richness ($S^* = 99.5$)
- **Far Small islands** ($D = 500$ km, $A = 10$ km²) support fewest species ($S^* = 96.8$)
- Turnover rates vary inversely with equilibrium richness: isolated, small islands show high turnover relative to their low diversity

Remark 5.1 (Species-Area Scaling) The fitted exponent $z = 0.243$ falls within the expected range for continental islands ($0.20 < z < 0.35$), confirming that larger islands support disproportionately more species through reduced extinction rates and increased habitat diversity. The power-law form implies a 10-fold increase in area yields approximately $10^{0.26} \approx 1.8$ -fold increase in species richness.

5.1 Colonization and Relaxation Dynamics

Example 5.2 (Extinction Debt) Following habitat fragmentation that reduces area from 1000 to 100.0 km² and increases isolation from 50 to 250 km, species richness declines from $S = 99.5$ to a new equilibrium $S^* = 98.7$ over approximately 100 years. This extinction debt represents species temporarily persisting above carrying capacity, gradually succumbing to increased extinction pressure.

5.2 Conservation Implications: The SLOSS Debate

Remark 5.2 (Reserve Design Trade-offs) The SLOSS analysis reveals:

- A single large reserve (1000 km²) supports $S = 99.4$ species
- Ten small reserves (100 km² each) collectively support $S \approx 365.9$ species
- The optimal strategy depends on: (1) target taxa dispersal ability, (2) habitat heterogeneity captured by multiple sites, (3) metapopulation rescue effects, (4) vulnerability to catastrophic disturbance

Modern consensus favors large reserves for area-sensitive species (e.g., top predators) while recognizing that multiple reserves can protect greater habitat diversity and provide insurance against localized extinctions.

6 Conclusions

This computational analysis demonstrates the MacArthur-Wilson Theory of Island Biogeography as a quantitative framework for understanding and predicting species richness patterns:

1. Equilibrium species richness emerges from the balance between immigration and extinction, with near large islands supporting $S^* = 99.5$ species compared to $S^* = 96.8$ for far small islands
2. The species-area relationship $S = cA^z$ yields a fitted exponent $z = 0.243$, consistent with continental island biotas ($R^2 = 0.964$)
3. Colonization dynamics show asymptotic approach to equilibrium over timescales of decades to centuries
4. Habitat fragmentation creates extinction debt, with species richness relaxing to lower equilibrium values as isolation increases and area decreases
5. SLOSS analysis indicates that reserve design must balance single-reserve area effects against among-reserve complementarity and risk-spreading

The equilibrium theory provides essential insights for conservation biology, particularly in designing nature reserve networks and predicting biodiversity responses to habitat fragmentation and climate-driven range shifts.

Further Reading

- MacArthur, R.H. & Wilson, E.O. *The Theory of Island Biogeography*. Princeton University Press, 1967.
- Lomolino, M.V. “Ecology’s most general, yet protean pattern: the species-area relationship.” *Journal of Biogeography* 27:17–26 (2000).
- Diamond, J.M. “The island dilemma: lessons of modern biogeographic studies for the design of natural reserves.” *Biological Conservation* 7:129–146 (1975).
- Whittaker, R.J. & Fernández-Palacios, J.M. *Island Biogeography: Ecology, Evolution, and Conservation*, 2nd ed. Oxford University Press, 2007.
- Hanski, I. “Metapopulation dynamics.” *Nature* 396:41–49 (1998).
- Rosenzweig, M.L. *Species Diversity in Space and Time*. Cambridge University Press, 1995.
- Simberloff, D. & Abele, L.G. “Island biogeography theory and conservation practice.” *Science* 191:285–286 (1976).
- Tjørve, E. “Shapes and functions of species-area curves: a review of possible models.” *Journal of Biogeography* 30:827–835 (2003).
- Losos, J.B. & Ricklefs, R.E. (eds.) *The Theory of Island Biogeography Revisited*. Princeton University Press, 2010.
- Kohn, D.D. & Walsh, D.M. “Plant species richness — the effect of island size and habitat diversity.” *Journal of Ecology* 82:367–377 (1994).
- Brooks, T.M. et al. “Habitat loss and extinction in the hotspots of biodiversity.” *Conservation Biology* 16:909–923 (2002).

- Holt, R.D. “On the evolutionary ecology of species’ ranges.” *Evolutionary Ecology Research* 5:159–178 (2003).
- Hubbell, S.P. *The Unified Neutral Theory of Biodiversity and Biogeography*. Princeton University Press, 2001.
- Warren, B.H. et al. “Islands as model systems in ecology and evolution: prospects fifty years after MacArthur-Wilson.” *Ecology Letters* 18:200–217 (2015).
- Pimm, S.L. et al. “The biodiversity of species and their rates of extinction, distribution, and protection.” *Science* 344:1246752 (2014).
- Fahrig, L. “Effects of habitat fragmentation on biodiversity.” *Annual Review of Ecology, Evolution, and Systematics* 34:487–515 (2003).
- Triantis, K.A. et al. “The island species-area relationship: biology and statistics.” *Journal of Biogeography* 39:215–231 (2012).
- Hanski, I. & Gaggiotti, O.E. (eds.) *Ecology, Genetics and Evolution of Metapopulations*. Academic Press, 2004.
- Schoener, T.W. “The MacArthur-Wilson equilibrium model: A chronicle of what it said and how it was tested.” In Losos & Ricklefs, 2010.
- Wilcox, B.A. & Murphy, D.D. “Conservation strategy: the effects of fragmentation on extinction.” *American Naturalist* 125:879–887 (1985).